Erdős Problem 848: A Kernel-Checked Proof of the Exact Extremal Bound

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Abstract

For $N \geq 1$, let $A \subseteq [1, N]$ satisfy the condition that $ab + 1$ is nonsquarefree for every $a, b \in A$. We determine the exact maximum cardinality of such a set, proving that

$$|A| \leq \left|\{1 \leq n \leq N : n \equiv 7 \pmod{25}\}\right|,$$

with equality attained by the progression $7 \pmod{25}$. Thus the extremal value is established for every N, closing the finite range left by the preceding asymptotic results. The proof begins with an exact Hall reformulation. A prefix-colouring certificate handles $N \leq 5 \cdot 10^6$; for larger N, uniform square-divisor estimates, opposite-base matchings, and valuation–cell–fibre decompositions reduce the remaining Hall defects to finitely many exact rational certificate states, together with a uniform argument on the unbounded tail. Certificate generators lie outside the trusted base: they emit ordinary Lean data, semantic checkers prove the required statements about that data, and the Lean 4 kernel replays the terminal all-N theorem with `--trust=0`. The complete proof archive, semantic checkers, theorem map, and reproduction metadata are available at Zenodo, concept DOI: 10.5281/zenodo.21750213.

1 Introduction

Write

$$C_N = \{n \in [1, N] : n \equiv 7 \pmod{25}\}.$$

Erdős and Sárközy asked whether this construction is extremal [3]. Sawhney proved the assertion for all sufficiently large N by a density sieve and structural reduction [11]. Sothanaphan made the asymptotic inputs explicit and obtained the threshold $N \geq 2.64 \cdot 10^{17}$ [12]. We prove the exact statement

$$|A| \leq |C_N| \tag{1}$$

for every positive integer N.

The computational part is not a pointwise enumeration of N. The Hall reformulation replaces the extremal-set question by a neighbourhood inequality. Monotone rational bounds compress long ranges to finitely many rows. Valuation, residue-cell, and prime-square-fibre partitions compress the small tail ranges to finitely many structural states. Only these states are certified. The bounded remainder is discharged by independently checkable arithmetic data, not by extrapolation from numerical evidence [6, 8].

Work	Proven range and conclusion	Remaining obstruction	Mechanism used here
Sawhney (2025)	The class 7 mod 25 is extremal for every $N \geq N_0$, for some unspecified N_0; a stability statement identifies the two candidate extremal classes.	The argument is asymptotic and gives neither N_0 nor the finite prefix.	Density sieves and the mod-25 structural reduction.
Sothanaphan (2026)	The same extremal bound holds explicitly for $N \geq 2.64 \cdot 10^{17}$.	The interval below that threshold remains; a pointwise search would not be a practical proof.	Explicit squarefree-progression estimates and a Pell-equation bound for large square divisors.
This paper	The exact extremal value is proved for every $N \geq 1$.	No range remains.	Hall reformulation, range compression, valuation–cell–fibre descent, and kernel-certified finite states.

Table 1: Prior results and the all-N theorem.

The squarefree-progression estimates belong to the classical line of Prachar and Hooley [9, 7]. Here they are combined with exact residue and valuation information tailored to the Hall graph. The machine proof follows the certificate-checking separation used in major formalized computational proofs [4, 5].

Proposition 1.1 (Sharpness)**.** *The set C_N is admissible: if $a, b \in C_N$, then $ab + 1$ is not squarefree.*

Proof. Since $a \equiv b \equiv 7 \pmod{25}$,

$$ab + 1 \equiv 7^2 + 1 = 50 \equiv 0 \pmod{25}.$$

Thus $5^2 \mid ab + 1$. □

2 The exact Hall cut

Throughout, N is a positive integer and $[1, N] = \{1, \ldots, N\}$. Join $b \in [1, N] \setminus C_N$ to $a \in C_N$ when $ab + 1$ is squarefree. For $B \subseteq [1, N] \setminus C_N$, put

$$\Gamma_N(B) = \{a \in C_N : ab + 1 \text{ is squarefree for some } b \in B\}$$

and

$$T_N(B) = C_N \setminus \Gamma_N(B).$$

Thus $a \in T_N(B)$ exactly when $ab+1$ is nonsquarefree for every $b \in B$. For every B whose internal products plus one are nonsquarefree, the Hall inequality is

$$|B| + |T_N(B)| \leq |C_N|. \tag{2}$$

Theorem 2.1 (Hall reduction)**.** *If (2) holds for every compatible outside set B, then (1) holds.*

Proof. Let A be admissible and put $B = A \setminus C_N$. Then $A \cap C_N \subseteq T_N(B)$, whence

$$|A| = |B| + |A \cap C_N| \le |B| + |T_N(B)| \le |C_N|.$$

$\square$

Theorem 2.2 (Hall equivalence)**.** *The Hall inequality* (2) *is equivalent to the exact upper bound* (1)*.*

Proof. The forward implication is Theorem 2.1. Conversely, assume (1) and let B satisfy the hypotheses of (2). Put $A = B \cup T_N(B)$. Products internal to B are nonsquarefree by hypothesis; products internal to $T_N(B)$ are divisible by 25; and every cross-product is nonsquarefree by the definition of $T_N(B)$. Hence A is admissible. The two parts are disjoint, so

$$|B| + |T_N(B)| = |A| \le |C_N|.$$

$\square$

For later normalization define the Hall completion

$$H_N(B) = B \cup T_N(B).$$

The union is disjoint, so $|H_N(B)| = |B| + |T_N(B)|$. The exact lower bound

$$|C_N| \ge \frac{N}{25} - \frac{7}{25} \tag{3}$$

shows that

$$\frac{|H_N(B)|}{N} \le \frac{1}{25} - \frac{7}{25N}$$

implies (2). All rational certificate rows below are checked against this target. We call a compatible outside set with $|C_N| < |B| + |T_N(B)|$ a *strict Hall defect*.

3 The exact five-million prefix

Theorem 3.1 (Exact prefix colouring)**.** *For every* $1 \le N \le 5 \cdot 10^6$*, let*

$$\mathcal{D}_N = \{x \in [1, N] : x^2 + 1 \text{ is nonsquarefree}\},$$

and join distinct $x, y \in \mathcal{D}_N$ *when* $xy + 1$ *is nonsquarefree. This graph has a proper colouring whose colour set is indexed by* C_N*. Consequently* (1) *and* (2) *hold throughout this interval.*

Proof. Every admissible set is a clique in this graph, so the colouring implies the cardinality bound. It remains to describe the checked certificate.

Put $L = 5 \cdot 10^6$. At a prefix write

$$U = \{u \le N : u \equiv 7 \pmod{25}\}, \qquad V = \{v \le N : v \equiv 18 \pmod{25}\}, \qquad E = \mathcal{D}_N \setminus (U \cup V).$$

For every $u \le L$ in the 7 mod 25 class, the certificate stores a chain of half-open intervals partitioning $[u, L + 1)$. On each interval its anchor state contains u, at most one member of V, and at most one member of E. The semantic checker verifies the interval chain, the three role constraints, and squarefreeness of every product between two occupied roles.

There are 200000 anchor histories and 200000 histories for the 18 mod 25 vertices. There are 125808 remaining diagonal histories. Each candidate history is a contiguous interval partition through $L+1$ and assigns the candidate to one checked anchor state whenever it is present. The outside diagonal marker is also checked in Lean. If $x \in E$ and x^2+1 is nonsquarefree, some prime square p^2 with $p \neq 5$ divides it; necessarily p is odd and $p \equiv 1 \pmod 4$. The verified roots of $z^2 \equiv -1 \pmod{p^2}$ place x in the marker. Conversely every marked progression carries its prime-square witness. Thus every vertex receives exactly one active colour at every prefix, and two distinct vertices with the same colour have squarefree product plus one. The anchor set at N is precisely C_N. □

4 Uniform tail reductions from five million

Let

$$D_N = \{n \in [1,N] : n \equiv 18 \pmod{25}\}.$$

For a Hall completion $H_N(B)$ put

$$R_N(B) = H_N(B) \setminus (C_N \cup D_N).$$

The first task is to eliminate a defect supported only on D_N; the second is to quantify the residual once a genuinely mixed pivot exists.

4.1 Pure opposite-base matching

For each parity, split the source D_N into that parity and the target C_N into the opposite parity. The target block has at least the size of the source block: the injection $x \mapsto x-11$ preserves the mod-25 classes and reverses parity. For opposite parities, $xy+1$ is odd, so the square prime 2 is inactive; moreover $xy+1 \equiv 18 \cdot 7 + 1 \equiv 2 \pmod{25}$, so 5 is inactive.

For a fixed pivot, nonsquarefree values are divided into three disjoint prime ranges:

$$p \le 47, \qquad 47 < p \le N/26, \qquad p > N/26.$$

The following three lemmas are cardinality bounds for the bad-point sets used in Lean.

Lemma 4.1 (Small-prime payment)**.** *Let $N \ge 5 \cdot 10^6$, let $X \subseteq [1,N]$ lie in one residue class modulo* 50*, and let b be any pivot. Then*

$$\frac{\#\{x \in X : \exists p \in \{3,7,11,13,17,19,23,29,31,37,41,43,47\},\ p^2 \mid bx+1\}}{N} \le \frac{S_0}{5 \cdot 10^6},$$

where

$$S_0 = 100001 \sum_p \frac{1}{p^2} + 13$$

and the sum is over the displayed thirteen primes.

Proof. For a fixed p, the conditions $x \bmod 50$ and $p^2 \mid bx+1$ select at most one class modulo $50p^2$, hence at most

$$\left\lfloor \frac{N+1}{50p^2} \right\rfloor + 1$$

points. Sum this bound over the thirteen primes. After division by N, the endpoint term is decreasing for $N \ge 5 \cdot 10^6$, so its maximum is the displayed value at $5 \cdot 10^6$. □

Lemma 4.2 (Medium-prime payment)**.** *Under the same hypotheses, the contribution of* $47 < p \le N/26$ *is at most*

$$\frac{3887}{25\cdot 10^6}+\frac{59}{10^4}.$$

Proof. The residue-class union bound gives the reciprocal-square contribution

$$\frac{1}{25}\sum_{47<p\le N/26}\frac{1}{p^2}.$$

The certified prime-square tail is less than $3887/10^6$. The terminal class contributes $\pi(N/26)/N$, and an explicit prime-count estimate bounds this by $59/10^4$ for $N \ge 5\cdot 10^6$ [10, 2]. Adding the two terms proves the claim. □

Lemma 4.3 (High-prime payment)**.** *Assume in addition that* $1 \le b \le N$ *and that* $bx+1$ *is coprime to* 10 *for every* $x \in X$*. Then the contribution of primes* $p > N/26$ *is at most*

$$\frac{11264064/3125}{5\cdot 10^6}.$$

Proof. Write $bx+1 = p^2 m$. The split $p > N/26$ gives $m \le 676$. After writing the fixed mod-50 progression parameter, the congruence becomes a transformed root problem. The support screen records the distinct prime divisors of b, and the two nonzero square cosets modulo 5 are treated separately. For every support prefix, the generated word mask is checked pointwise against its quadratic-residue predicate. The transformed-parameter cardinality theorem then bounds the row by its stored rational root coefficient. The support-product theorem proves that every possible pivot selects a checked row. The largest checked coefficient is 11264064/3125, and division by N is maximized at $N = 5\cdot 10^6$. □

Theorem 4.4 (Uniform pure opposite-base matching)**.** *For every* $N \ge 5\cdot 10^6$*, the squarefree-edge graph from* D_N *to* C_N *has a matching saturating* D_N*. Consequently every compatible* $B \subseteq D_N$ *satisfies* (2)*.*

Proof. The exact rational calculation in Lean proves

$$\frac{S_0}{5\cdot 10^6}+\left(\frac{3887}{25\cdot 10^6}+\frac{59}{10^4}\right)+\frac{11264064/3125}{5\cdot 10^6}\le\frac{1}{100}-\frac{1}{5\cdot 10^6}. \tag{4}$$

Lemmas 4.1–4.3 therefore show, in each parity block and in both directions, that strictly more than half of the opposite block consists of squarefree neighbours. The elementary dense bipartite matching lemma now applies: the source side has size at most the target side, and every vertex has degree greater than half of the opposite side. Hall's condition follows by considering subsets of size at most and greater than half separately. The two parity matchings combine because their target parities are disjoint. Finally, a matching from $B \subseteq D_N$ into $C_N \setminus T_N(B)$ gives $|B| \le |C_N| - |T_N(B)|$. □

4.2 A quantitative mixed residual

Proposition 4.5 (Uniform mixed degree)**.** *Let* $N \ge 5\cdot 10^6$*, let* B *be a compatible outside set, and let* $b \in R_N(B)$*. Then* b *has more than*

$$\frac{N}{50}+\frac{N}{525}$$

squarefree neighbours in each of C_N *and* D_N*.*

Proof. Because $b \in R_N(B)$, the square 5^2 cannot divide the relevant forms. Split a nonsquarefree form $bx + 1$ into the three ranges

$$p \in \{2, 3, 7\}, \qquad 7 < p \leq N/55, \qquad p > N/55.$$

Put

$$\tau_7 = \frac{2635290839090428865173764611843337967}{857345679663769237990628978784100000},\qquad \rho_7 = \frac{221926420176}{12755647965025}.$$

The first part is counted by the exact periodic masks and has normalized envelope $17/1225$. The second is bounded by the certified reciprocal-square tail $\tau_7/25$. In the last range the all-support transformed-root rows, including all three even valuations, give $\rho_7/6$. The base progression contains at least $N/25 - 1$ points. The exact margin is

$$\frac{17}{1225} + \frac{\tau_7}{25} + \frac{\rho_7}{6} + \frac{1 + 19601/49}{5 \cdot 10^6} < \frac{1}{50} - \frac{1}{525}.$$

The endpoint correction decreases with N, so subtracting all bad points from the base progression gives the stated degree in both directions. $\square$

Corollary 4.6 (Uniform mixed residual)**.** *If $N \geq 5 \cdot 10^6$ and a compatible outside set B violates* (2)*, then*

$$\frac{2N}{525} - 1 < |R_N(B)|. \tag{5}$$

In particular $|R_N(B)| > 19046$, and one of the five valuation parts used below has more than 3809 elements.

Proof. Theorem 4.4 implies that $R_N(B)$ is nonempty; choose $b \in R_N(B)$. Apply Proposition 4.5 to both base classes. The exact degree-subtraction identity removes the neighbours already occupied by B and the non-neighbours $T_N(B)$ and gives

$$\frac{2N}{525} - 1 < |R_N(B)|.$$

At $N = 5 \cdot 10^6$ the left side is greater than 19046, and it increases with N. The five valuation parts are disjoint and cover $R_N(B)$. If each had size at most 3809, their union would have size at most $5 \cdot 3809 = 19045$, a contradiction. $\square$

5 Valuation, cell, and fibre descent

The five valuation classes are

$$E_1,\ E_2,\ E_3,\ O_1,\ O_3,$$

where, respectively,

$$x \equiv 2 \pmod 4, \quad x \equiv 4 \pmod 8, \quad x \equiv 0 \pmod 8, \quad x \equiv 1 \pmod 4, \quad x \equiv 3 \pmod 4.$$

Within a class we next split by the nine residues modulo 9; within a cell we use prime-square fibres, principally modulo 7^2 and 11^2.

Lemma 5.1 (Valuation partition)**.** *The five valuation parts are pairwise disjoint and their union is $R_N(B)$. Consequently their cardinalities sum to $|R_N(B)|$.*

Proof. Each integer has exactly one of the five defining valuation patterns. The formal proof reduces the five cases to parity and exact 2-adic divisibilities, proves pairwise disjointness, and applies cardinality of a disjoint finite union. $\square$

Lemma 5.2 (Nine-cell capacity dichotomy)**.** *Fix one odd valuation class after the previously charged points have been removed, and let $\mathcal{R}$ be the resulting structured residual. Join a residue label $c \pmod 9$ to a residue $r \pmod{49}$ when some $x \in \mathcal{R}$ satisfies $x \equiv c \pmod 9$ and $x \equiv r \pmod{49}$. For a set Y of mod-9 labels, let $\Gamma_{49}(Y)$ be its neighbourhood in the mod-49 residues. Then either eight distinct mod-9 labels can be assigned occupied mod-49 fibres, with no fibre used more than twice, or there is a set Y for which*

$$2\,|\Gamma_{49}(Y)| + 1 < |Y|.$$

Proof. Apply Hall's theorem to the bipartite graph whose left side is the nine mod-9 labels and whose right side consists of two copies of every mod-49 residue together with one spare vertex. A saturating matching uses the spare at most once. Every other matched edge has, by definition, an occupant in $\mathcal{R}$; choosing one produces eight pivots in distinct mod-9 cells, with each mod-49 residue used at most twice. If no saturating matching exists, Hall supplies Y with the displayed capacity defect. $\square$

For every terminal branch the checker proves an inequality of the form

$$\frac{|H_N(B)|}{N} \le D_\beta + F_\beta + Q_\beta < \frac{1}{25} - \frac{7}{25L} \le \frac{1}{25} - \frac{7}{25N}. \tag{6}$$

Here L is the lower endpoint, D_β is the diagonal/residual payment, F_β is an exact finite-prime payment, and Q_β is the reciprocal-square plus transformed-root payment. The first inequality is a semantic theorem about the selected pivots; the middle inequality is exact rational arithmetic; the last is monotonicity. Thus (6) contradicts a strict Hall defect.

Theorem 5.3 (The five-to-ten-million range)**.** *The Hall inequality holds for*

$$5 \cdot 10^6 \le N < 10^7.$$

Proof. Assume a strict defect. Corollary 4.6 gives $|R_N(B)| > 19046$, in particular more than the 128 points needed after all finite charges. The case analysis examines E_1, then E_2, then E_3, then a one-odd-class allocation, and finally the remaining two-odd-class case. At each step either the cell and fibre lemmas construct a terminal pivot pattern, or the entire failed class has the exact bounded cardinality charged to the next step. The last step cannot fail because the five valuation parts cover $R_N(B)$.

Every terminal object contains its selected pivots, the proof that they lie in $H_N(B)$, the exact finite-prime bound, the transformed-root bound, and the diagonal bound. Its semantic theorem yields the first inequality in (6); its checked rational budget yields the strict middle inequality with $L = 5 \cdot 10^6$. This contradicts the defect. $\square$

Theorem 5.4 (The ten-to-twenty-million range)**.** *The Hall inequality holds for*

$$10^7 \le N < 2 \cdot 10^7.$$

Proof. Again $|R_N(B)| > 19046$. In E_1 and then E_2, the case analysis tests for two dense mod-9 cells, then one dense cell, and otherwise charges all nine cells. It next tests E_3. If all three even classes are charged, their total charge is explicit and at least 91 residual pivots remain in the two odd classes. The odd case analysis then returns a three-pivot terminal. The two-dense-cell alternatives return four-pivot terminals; the one-cell and odd alternatives return three-pivot terminals. These cases are exhaustive by construction.

Both terminal types prove their actual Hall-completion ratio strictly below $1/25 - 7/(25N)$. The proof uses the checked periodic finite-prime rows, root bounds, diagonal bounds, and exact cell charges. Hence a strict defect is impossible. □

Theorem 5.5 (The twenty-to-forty-million range). *The Hall inequality holds for*

$$2 \cdot 10^7 \leq N < 4 \cdot 10^7.$$

Proof. The cutoff-19 allocation theorem selects one of ten branches: generic or common-mod-9 triples in each of E_1, E_2, E_3, followed by generic or common-mod-9 branches in two odd classes or in one odd class. The even branches use finite cutoff 23; the odd branches use cutoff 19. For a terminal certificate with branch β, Lean proves

$$\frac{|H_N(B)|}{N} \leq T_\beta,$$

where T_β is the sum of the certified residual, finite-prime, and tail payments. Exact rational normalization proves

$$T_\beta < \frac{1}{25} - \frac{7}{252 \cdot 10^7} \leq \frac{1}{25} - \frac{7}{25N}$$

for all ten branches. The selected branch therefore contradicts the assumed Hall defect. □

6 The compressed middle ranges

Lemma 6.1 (Ten-branch allocation). *Let $N \geq 2 \cdot 10^7$ and suppose that B is a compatible outside set with a strict Hall defect. Then one of the ten branches used in the proof of Theorem 5.5 applies.*

Proof. The uniform cutoff-19 degree theorem gives the required residual mass. Put $g_N = \lfloor (N-1)/20001 \rfloor + 1$ and inspect E_1, E_2, E_3 in order. If the current class has more than g_N members, a three-pivot gap selection either is nonconstant modulo 9 or lies in one mod-9 cell. Otherwise charge the class by at most g_N. If all three are charged, the remaining mass lies in the two odd classes; the same gap selection and mod-9 dichotomy gives the final four branches. Each unsuccessful test has the cardinality bound used by the next test, so the alternatives are exhaustive. □

Theorem 6.2 (The forty-to-two-hundred-million range). *The Hall inequality holds for*

$$4 \cdot 10^7 \leq N < 2 \cdot 10^8.$$

Proof. The interval is divided into the six half-open blocks with endpoints

$$40,\ 50,\ 70,\ 80,\ 100,\ 150,\ 200 \quad \text{million}.$$

Lemma 6.1 selects a branch. On each block, the residual payment is a checked diagonal envelope plus the exact charges from preceding sparse classes. The finite-prime payments, in branch order, are

$$10^{-6}(8685, 12616, 8685, 12616, 8685, 12616, 19420, 20878, 26643, 29459).$$

The tail payment uses cutoff 23 in the six even branches and cutoff 19 in the four odd branches, together with the four blockwise transformed-root envelopes. The semantic terminal theorem gives the first inequality in (6). The checked rational table verifies all $6 \cdot 10$ comparisons, and block membership gives the final monotonicity step. Hence no strict defect exists. □

Theorem 6.3 (The two-hundred-million-to-two-billion range)**.** *The Hall inequality holds for*

$$2 \cdot 10^8 \leq N < 2 \cdot 10^9.$$

Proof. The four structural regimes have endpoints

$$200,\ 300,\ 500,\ 1000,\ 2000 \quad \text{million}$$

and transformed-root splits $75, 85, 100, 125$, respectively. Their quotient boxes have sizes $75^2, 85^2, 100^2, 125^2$. The normal and mod-5-twisted word tables certify the quadratic-residue survivor counts for every support length through eight. A separate finite list of root-envelope rows covers the four regimes, and its cover checker proves that every integer in those regimes lies in the domain of a certified row.

Lemma 6.1 supplies one of the same ten branches. For a regime ρ and branch β, define

$$T_{\rho,\beta} = D_{\rho,\beta} + F_\beta + Q_{\rho,\beta},$$

where the diagonal envelope $D_{\rho,\beta}$ is selected from the eight checked diagonal classes, F_β is the finite-prime vector displayed in the preceding proof, and $Q_{\rho,\beta}$ uses the regime root ceiling

$$10^{-9}(9000000,\ 7700000,\ 6400000,\ 5000000).$$

The root certificate consists only of finite word masks and checked row data; it asserts no Hall inequality. The semantic checker derives the actual transformed-root inequality from those data. The terminal theorem then proves $|H_N(B)|/N \leq T_{\rho,\beta}$, and exact rational arithmetic proves

$$T_{\rho,\beta} < \frac{1}{25} - \frac{7}{25L_\rho} \leq \frac{1}{25} - \frac{7}{25N}$$

for all $4 \cdot 10$ cases. This contradicts a strict defect. □

7 The high tail

The finite high tail uses six complete rows. Each row contains a checked diagonal budget, four checked quadratic-residue root rows (for E_1, E_2, E_3 and the odd class), endpoint-alignment proofs, and the Boolean proof that all ten branch totals are below the target.

Theorem 7.1 (Finite high-tail certificate)**.** *The six certified rows prove the Hall inequality for*

$$2 \cdot 10^9 \leq N < 5 \cdot 10^{11}.$$

Proof. The half-open row endpoints are

$$2,\ 2.3,\ 2.8,\ 4.2,\ 6,\ 12,\ 500 \quad \text{billion}.$$

Their prime profiles are respectively $16, 18, 18, 18, 30, 30$. The list-level cover predicate checks that the first lower endpoint is 2 billion, each next lower endpoint is one more than the previous inclusive upper endpoint, and the stop is 500 billion. Hence every N in the displayed range selects a unique complete row. Lemma 6.1 selects a branch, and the row's semantic theorem bounds the actual Hall completion by that branch total. The stored Boolean equality is converted to the proposition that every branch total is strictly below $1/25 - 7/(25N)$. Thus (2) follows. □

Theorem 7.2 (Unbounded high tail)**.** *The Hall inequality holds for every*

$$N \geq 5 \cdot 10^{11}.$$

Proof. Use split 120, root floor 840, and the anchored 47-prime sieve. The common root envelope is

$$\frac{8062340}{10^9}.$$

The eight diagonal envelopes are

$$10^{-9}(26005710, 20944421, 17208292, 16649746, 13003280, 7437057, 6604680, 917036),$$

with grouped wheel divisors

$$(1821, 2892, 2627, 2295, 3643, 4590, 3643, 3643).$$

The root checker proves the transformed-root inequality, the diagonal checker proves the eight semantic diagonal inequalities, and the ten-branch checker proves every resulting total below the target at $5 \cdot 10^{11}$. All positive endpoint corrections decrease with N, so the same bounds apply for every larger N. Combining the selected branch from Lemma 6.1 with these three checked implications rules out a strict Hall defect. □

Theorem 7.3 (Certified high range)**.** *The Hall inequality, and hence the exact upper bound, holds for every $N \geq 2 \cdot 10^9$.*

Proof. Theorem 7.1 covers $2 \cdot 10^9 \leq N < 5 \cdot 10^{11}$, and Theorem 7.2 covers $N \geq 5 \cdot 10^{11}$. □

8 Assembly

Theorem 8.1 (The range from forty million onward)**.** *The Hall inequality, and hence the exact upper bound, holds for every $N \geq 4 \cdot 10^7$.*

Proof. Theorems 6.2, 6.3, and 7.3 cover, respectively,

$$[4 \cdot 10^7, 2 \cdot 10^8), \qquad [2 \cdot 10^8, 2 \cdot 10^9), \qquad [2 \cdot 10^9, \infty).$$

□

Theorem 8.2 (The range from five million onward)**.** *The Hall inequality, and hence the exact upper bound, holds for every $N \geq 5 \cdot 10^6$.*

Proof. Theorems 5.3, 5.4, and 5.5 cover

$$[5 \cdot 10^6, 10^7), \quad [10^7, 2 \cdot 10^7), \quad [2 \cdot 10^7, 4 \cdot 10^7),$$

and Theorem 8.1 covers the remaining unbounded interval. These four half-open intervals are pairwise disjoint and cover $[5 \cdot 10^6, \infty)$. □

Theorem 8.3 (Erdős Problem 848)**.** *For every integer $N \geq 1$, the maximum cardinality of a set $A \subseteq [1, N]$ for which $ab + 1$ is nonsquarefree for every $a, b \in A$ is*

$$|\{1 \leq n \leq N : n \equiv 7 \pmod{25}\}| .$$

Equivalently, the Hall inequality (2) *holds for every N.*

Proof. Theorem 3.1 gives the upper bound for $N \leq 5 \cdot 10^6$, and Theorem 8.2 gives it for $N \geq 5 \cdot 10^6$. Proposition 1.1 supplies equality. The Hall formulation is equivalent by Theorem 2.2. □

9 Machine-checked proof and verification artifact

The formal artifact uses Lean 4 and Mathlib [1, 13]. Its role is to check the exact finite arithmetic, the manuscript-to-Lean interfaces, and the final all-N assembly with a small trusted kernel.

Manuscript-to-Lean correspondence

The final upper-bound declaration is

`Erdos848.PaperGeneratedCertificateProvider.all_N.`

Together with `originalA7_has_property` and `erdos848HallStatement_iff_originalProblem`, it yields the sharpness and Hall-equivalence clauses of Theorem 8.3. The machine-readable endpoint map contains fifteen published Lean declarations. A separate result map binds every numbered result in this manuscript to exact Lean declaration names and source files. The accompanying verification script compares the manuscript labels with the result map, checks each declaration in its cited source, and confirms that every cited module lies in the import closure of the final upper-bound theorem. The correspondence is therefore checked at declaration level.

Certificate production and trust boundary

External C++ or Python programs may search for witnesses, compress ranges, and emit Lean source. They are not trusted and are not rerun by Lean. Generated data are ordinary Lean terms. Semantic checkers prove their residue, valuation, cell, fibre, coverage, and rational-budget properties. Only the resulting proof terms are accepted by the kernel. Precomputed witnesses therefore change proof-production time, not the logical trust boundary.

The transitive publication cone contains exactly 30,638 local Lean source modules. It excludes project axioms, `sorry`, `admit`, `native_decide`, `unsafe`, and external code. A `--trust=0` axiom audit of the published endpoints reports exactly

`[propext, Classical.choice, Quot.sound],`

the standard classical and quotient dependencies used by Lean and Mathlib.

Paper component	Principal Lean declarations
Problem, sharpness, and Hall equivalence	`OriginalProblem848Statement`, `originalA7_has_property`, `erdos848HallStatement_iff_originalProblem`
Exact prefix through $5 \cdot 10^6$	`erdos848_through_five_million`
Uniform pure matching and mixed residual	`pureGlobalOppositeBaseMatching`, `globalMixedHallResidual_cast_lower_of_defect`
The first three tail intervals	`erdos848FiveToTenMillionClose`, `erdos848TenToTwentyMillionClose`, `erdos848TwentyMillionClose_kernel`
The middle and high subranges	`erdos848FortyMillionClose_kernel`, `erdos848PaperTwoHundredMillionToTwoBillionClose_of_certificates`, `HighTailCloseCertificate.close`
Unified tail and final upper bound	`erdos848_paper_tail_close`, `PaperGeneratedCertificateProvider.all_N`

Table 2: Principal manuscript-to-Lean correspondences. The complete per-result map is machine-readable.

Reproduction and cached artifacts

The version-independent Zenodo concept DOI for the code archive is `10.5281/zenodo.21750213`; it resolves to the latest archived code release. The repository pins the Lean toolchain and Mathlib dependency graph. The proof contract and release manifest bind the TeX, PDF, source closure, toolchain, endpoint map, result map, numeric claims, bibliography evidence, and axiom policy. The release-state audit is

```
python -B scripts/check_proof_state.py --require-release-ready --audit-sources
```

and the manuscript-specific gates are

```
python -B scripts/verify_paper_lean_correspondence.py
python -B scripts/verify_paper_lean_numbers.py
python -B scripts/verify_reference_evidence.py
    --require-cited-coverage --require-entry-checks
```

Optional precompiled `.olean` shards are a hash-checked transport cache. They can shorten replay, but they are neither mathematical premises nor a substitute for kernel checking.

Disclosure and verification statement

Multiple artificial-intelligence tools and an author-developed research system were used in the research and preparation of this article. The author is responsible for the mathematical content and conclusions of the article. All results can also be independently verified through the accompanying kernel-checked machine-proof project; their formal validity is therefore subject to reproducible kernel verification rather than the author's assertion alone. Further details concerning the systems and their use will be disclosed separately.

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